GEORGETOWN UNIVERSITY · DSAN 6725 FINAL PROJECT

Agentic AI Nutritionist

A multi-agent meal-planning system that combines persistent user memory with grounded recipe retrieval and a deterministic audit layer.

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IN BRIEF A multi-agent system that produces **personalized, nutritionally targeted meal plans** through conversation — grounded in a 751-recipe corpus, with a deterministic audit layer outside the LLM.
Perfect on safety and structure; corpus and latency are the remaining limits.

1 The Problem

motivation

Manual trackers like MyFitnessPal **burden the user**. Generative LLMs **hallucinate the macros**.

695 *kcal* **Five leading LLMs** systematically *underestimated* daily energy vs. registered-dietitian plans.
Bilen et al., 2026 · n = 5 models

We explore a middle path: an **agentic architecture** that combines persistent memory with grounded retrieval — and **isolates safety from LLM inference**.

Related work · *Angara 2017, Abbas 2026, Guo 2025, Bilen 2026, Khamesian 2024, Papastratis 2024.*

2 The Corpus

budget bytes · 751

Scraped from **Budget Bytes**. Category labels route recipes to the right meal slot.

751	$1.22	35 *min*	213 *kcal*
RECIPES	MEDIAN COST / SERVING	MEDIAN PREP + COOK	MEAN KCAL / SERVING

CAVEAT Nutrition values were *LLM-estimated* — not USDA. Production would replace with a lab-validated dataset.

3 The Stack

why these

`claude-sonnet-4.6` ORCHESTRATOR · INTAKE	`claude-haiku-4.5` PLANNER · STRUCTURED
`strands-agent-sdk` AGENT FRAMEWORK	`mem0` PERSISTENT MEMORY
`sqlite` PERSONAL COOKBOOK	`streamlit` FRONTEND

Chosen to keep the *what the model is doing* legible — Strands over LangChain, SQLite over vector-store.

4 System Architecture

figure 1

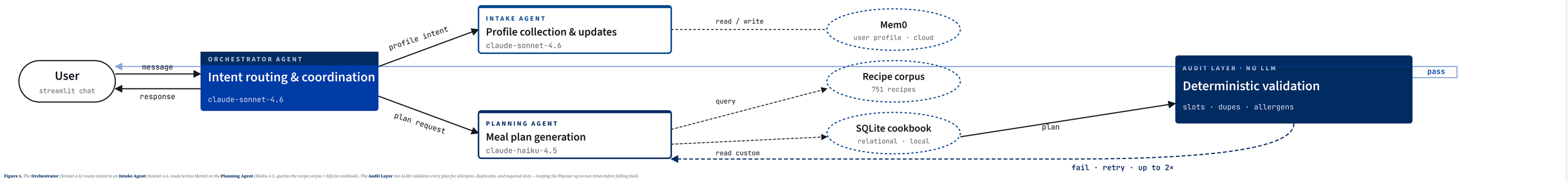

Figure 1. *The* **Orchestrator** *(Sonnet 4.6) routes intent to an* **Intake Agent** *(Sonnet 4.6, reads/writes Mem0) or the* **Planning Agent** *(Haiku 4.5, queries the recipe corpus + SQLite cookbook). The* **Audit Layer** *(no LLM) validates every plan for allergens, duplicates, and required slots — looping the Planner up to two times before falling back.*

5 The Safety Loop

no LLM in the validator

01 **Request** orchestrator → 02 **Compute targets** mifflin-st jeor → 03 **Query corpus** filter · score → 04 **Assemble** 1 or N days → *Audit?* slots · allergens → ✓ PASS **Present** to user

RETRY · 1: **Widen time**

RETRY · 2: **Widen ingredient count**

FALLBACK: **Best plan + notes**

6 How We Tested

braintrust · no LLM judge

29 CASES · 5 CATEGORIES

Meal Planning	12
Dietary Restriction	6
Profile Update	4
Out of Scope	4
Shopping List	3
Total	**29**

8 DETERMINISTIC SCORERS

- ToolSelection
- NoError
- PlanCompleteness
- AllergenSafety
- Latency
- ScopeAwareness
- CalorieTarget
- ShoppingList

Binary or banded numeric. CalorieTarget band: ±15% of Mifflin-St Jeor.

7 Results

figure 2 · table 1

FIG. 2 · AVG. SCORE BY SCORER

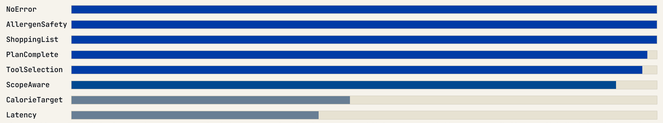

TABLE 1 · PER-CATEGORY (APP. B)

CATEGORY	TOOL	PLAN	CAL	ALLG	SCOPE	LAT
Meal Planning	0.983	0.986	0.518	–	1.000	0.208
Dietary Restr.	0.933	0.972	0.392	1.000	1.000	0.417
Shopping List	1.000	1.000	–	–	1.000	0.167
Profile Update	1.000	–	–	–	1.000	0.750
Out of Scope	–	–	–	–	0.500	0.938

Fig. 2 & Table 1. *Safety-critical and structural scorers are saturated. CalorieTarget (corpus-bound) and Latency (sequential inference) are unpacked in §8. Bar n = applicable cases; table bodies dashed where scorer does not apply.*

8 What It Means

findings

01 Grounding eliminates hallucination — by construction.
The system can't invent meals. The audit is string-matching.

02 Separated agents made debugging tractable.
Narrow prompts, predictable routing, isolated failures.

03 CalorieTarget (0.476) is a *corpus* problem.
Budget Bytes skews light (213 kcal/srv). USDA corpus closes most of the gap.

04 Latency (0.422) is an *infra* problem.
2–3 sequential API calls per turn. Parallelize + cache to fix.

9 Conclusion

TAKEAWAY

Agentic architecture delivers conversational UX *and* nutritional reliability — when meals come from a validated corpus and **the safety layer never touches an LLM**.

Next: USDA corpus · auth · grocery-MCP · temporal profile.

REPO · DEMO

& references

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Colophon. *Poster design iterated with Claude (Anthropic) via its design-tool workspace — layout, typography, and SVG figures were generated through conversation with the model. All authorial decisions, research, data, and system implementation are the authors' own.*